

Elementary Matrices and Row Equivalence

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Learning Outcomes

At the end of this lecture, you should be able to:

- Know what is the elementary matrices and their properties.
- Understand the Matrix Inversion Algorithm and know how to apply it.
- Know what is meant by one matrix is row equivalent to another.

Elementary matrices

An n -square matrix $E \in M_n(\mathbb{R})$ is called an **elementary matrix** if E can be obtained from the identity matrix I_n by performing a single elementary row operation.

All elementary matrices are invertible and their inverses are also elementary matrices.

[E1]: Interchanging the i^{th} row and the j^{th} row: $R_i \leftrightarrow R_j$

[E2]: Multiplication of a row by a nonzero constant λ : $R_i \rightarrow \lambda R_i$,
 $\lambda \in \mathbb{R} \setminus \{0\}$

[E3]: Addition of a constant multiple of one row to another:
 $R_i \rightarrow R_i + \beta R_j, \beta \in \mathbb{R}$

Elementary matrices

Elementary matrices

If a matrix B is obtained from A by performing a finite series of elementary row operations, then there exist elementary matrices $E_1, E_2, \dots, E_k$ such that

$$B = E_k \cdots E_2 E_1 A \text{ and } A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} B.$$

Example:

Suppose the matrix B is obtained from A by performing elementary row operations as below. For each of these elementary row operations, write down the corresponding elementary matrices E_k , $k = 1, 2, 3, 4$.

$$\begin{aligned} A = \begin{bmatrix} 0 & -3 & 5 \\ 2 & 5 & 3 \\ 1 & 2 & 3 \end{bmatrix} &\xrightarrow[E_1]{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 0 & -3 & 5 \end{bmatrix} \xrightarrow[E_2]{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & -3 & 5 \end{bmatrix} \\ &\xrightarrow[E_3]{R_3 \rightarrow R_3 + 3R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -4 \end{bmatrix} \xrightarrow[E_4]{R_3 \rightarrow -\frac{1}{4}R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} = B \end{aligned}$$

Elementary matrices

Elementary matrices

If $A \in M_n(\mathbb{R})$, then all of the following are equivalent:

- A is invertible.
- A is non-singular.
- The homogeneous linear system $AX = 0$ has only trivial solution.
- The linear system $AX = B$ has a unique solution for each $B \in M_{n,1}(\mathbb{R})$.
- The reduced row-echelon form of A is the identity matrix I_n .
- The matrix A can be expressed as a product of elementary matrices.

Matrix Inversion Algorithm

The Matrix Inversion Algorithm:

Let $A \in M_n(\mathbb{R})$ be an n -square matrix. If A can be transformed to the identity matrix I_n by a finite sequence of elementary row operations $O_1, O_2, \dots, O_k$, then A is invertible, and the same finite sequence of elementary row operations applied to yields the inverse matrix A^{-1} , i.e.,

$$A \xrightarrow{O_1, O_2, \dots, O_k} I_n \Rightarrow I_n \xrightarrow{O_1, O_2, \dots, O_k} A^{-1}$$

$$\text{or } \left[A \mid I_n \right] \xrightarrow{O_1, O_2, \dots, O_k} \left[I_n \mid A^{-1} \right].$$

Matrix Inversion Algorithm

Example:

Use the elementary row operations to determine the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

Matrix Inversion Algorithm

Recall that a square matrix is invertible if and only if its reduced row-echelon form is an identity matrix.

If the row-echelon form of a square matrix has no zero row, then it is invertible.

Otherwise, it is not invertible.

Example:

Determine whether the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 4 \\ 3 & 6 & 3 \end{bmatrix}$ is invertible.

Row equivalence

Let $A, B \in M_{m,n}(\mathbb{R})$ be two $m \times n$ matrices. We say that A is **row equivalent** to B , denoted by $A \stackrel{r}{\sim} B$, if B can be obtained from A by performing a finite sequence of elementary row operations.

For any $m \times n$ matrix A , $A \stackrel{r}{\sim} A$

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 4 \\ 3 & 6 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix}$. From previous example, B can be obtained from A by performing 3 elementary row operations. So $A \stackrel{r}{\sim} B$.

The row equivalence is an equivalence relation on $M_{m \times n}(\mathbb{R})$.

Row equivalence

If $A, B \in M_{m \times n}(\mathbb{R})$, then all of the following are equivalent:

- A is row equivalent to B .
- There exists elementary matrices $E_1, E_2, \dots, E_k \in M_m(\mathbb{R})$ such that $B = E_k E_{k-1} \cdots E_1 A$.
- There exists an invertible matrix $P \in M_m(\mathbb{R})$ such that $B = PA$.

Row equivalence

Let $A \in M_n(\mathbb{R})$. Then, A is invertible if and only if A and I_n are row equivalent.

Every matrix $A \in M_{m \times n}(\mathbb{R})$ is row equivalent to a row-echelon matrix.

Every matrix $A \in M_{m \times n}(\mathbb{R})$ has a unique reduced row-echelon matrix.